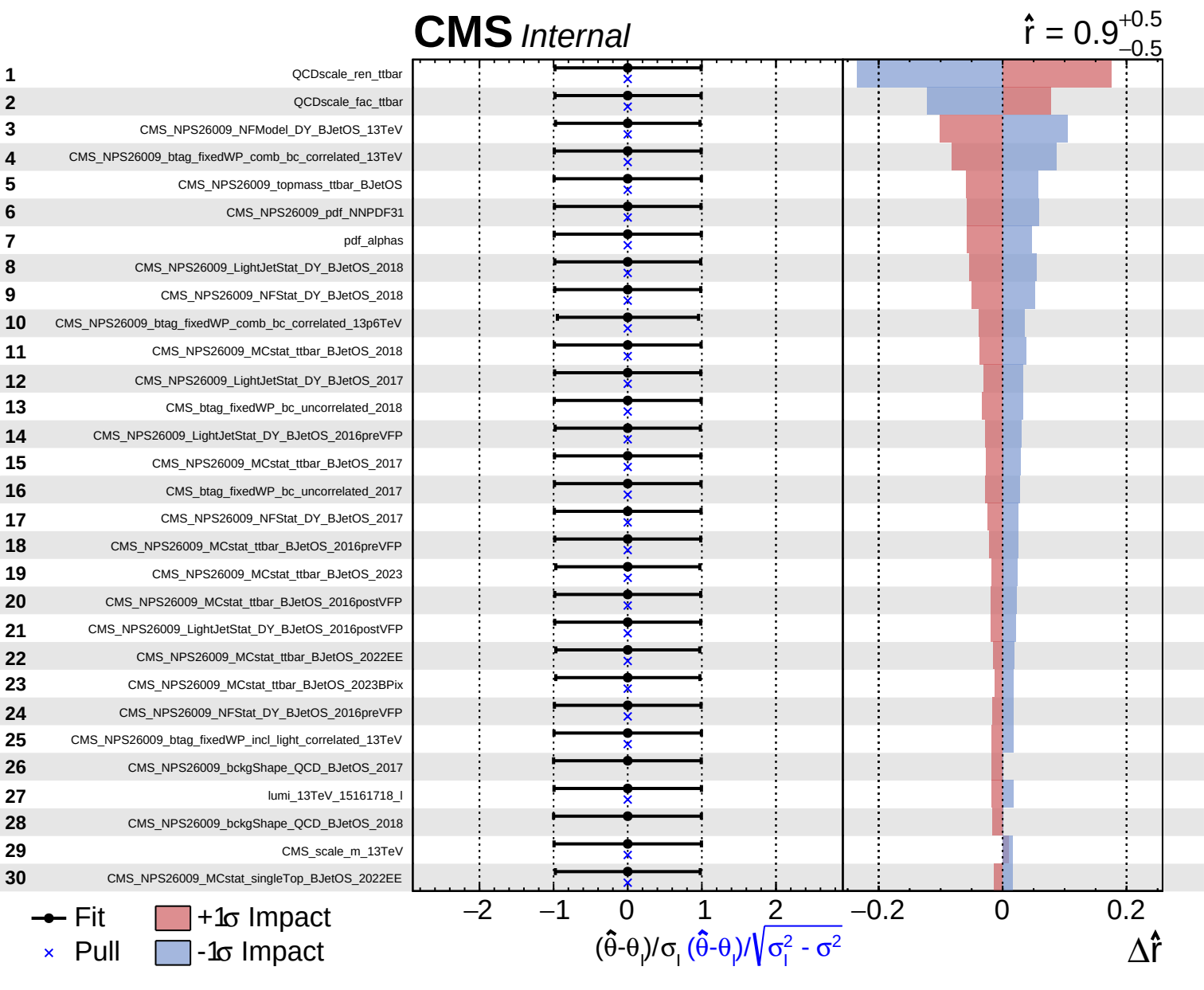
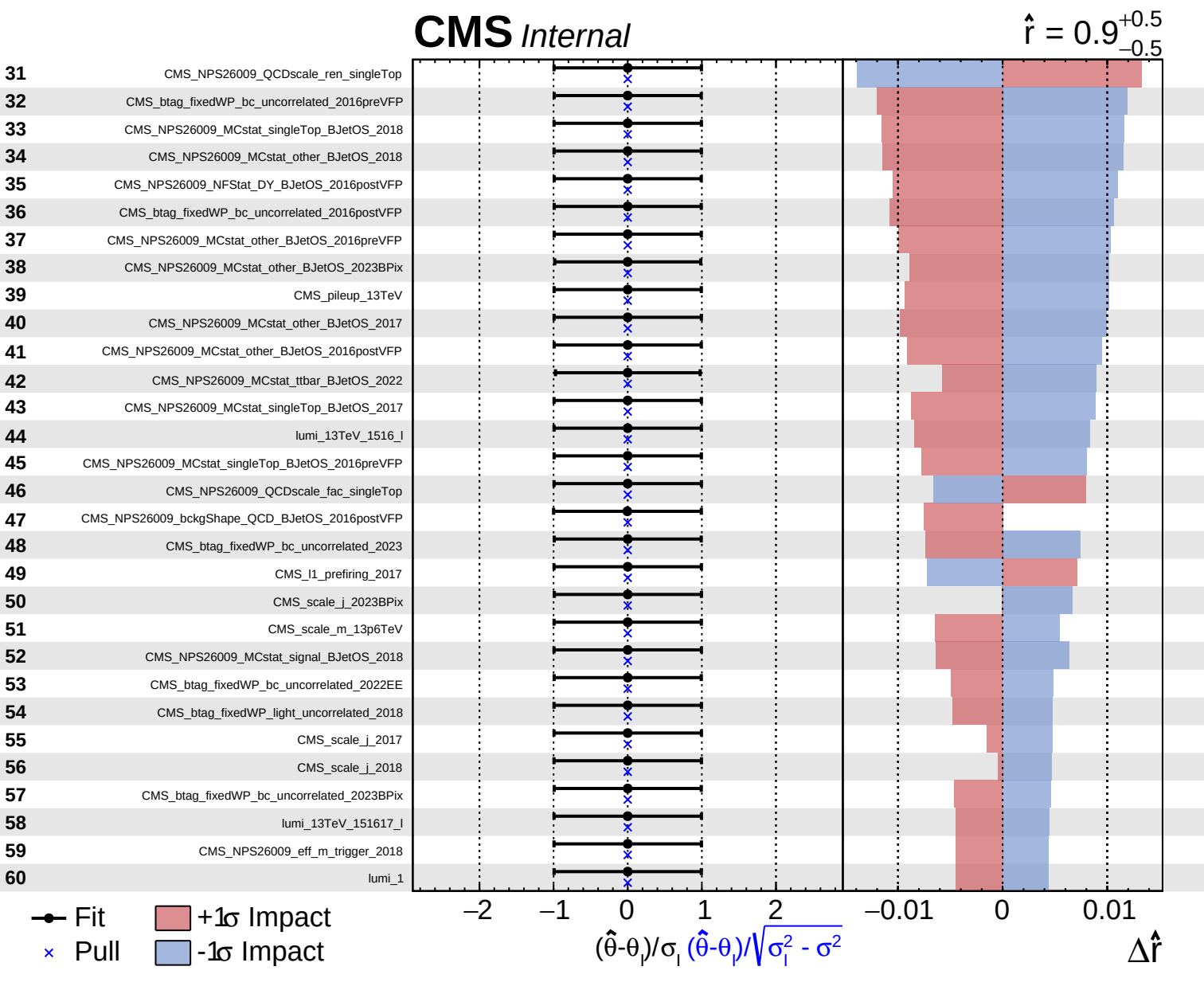
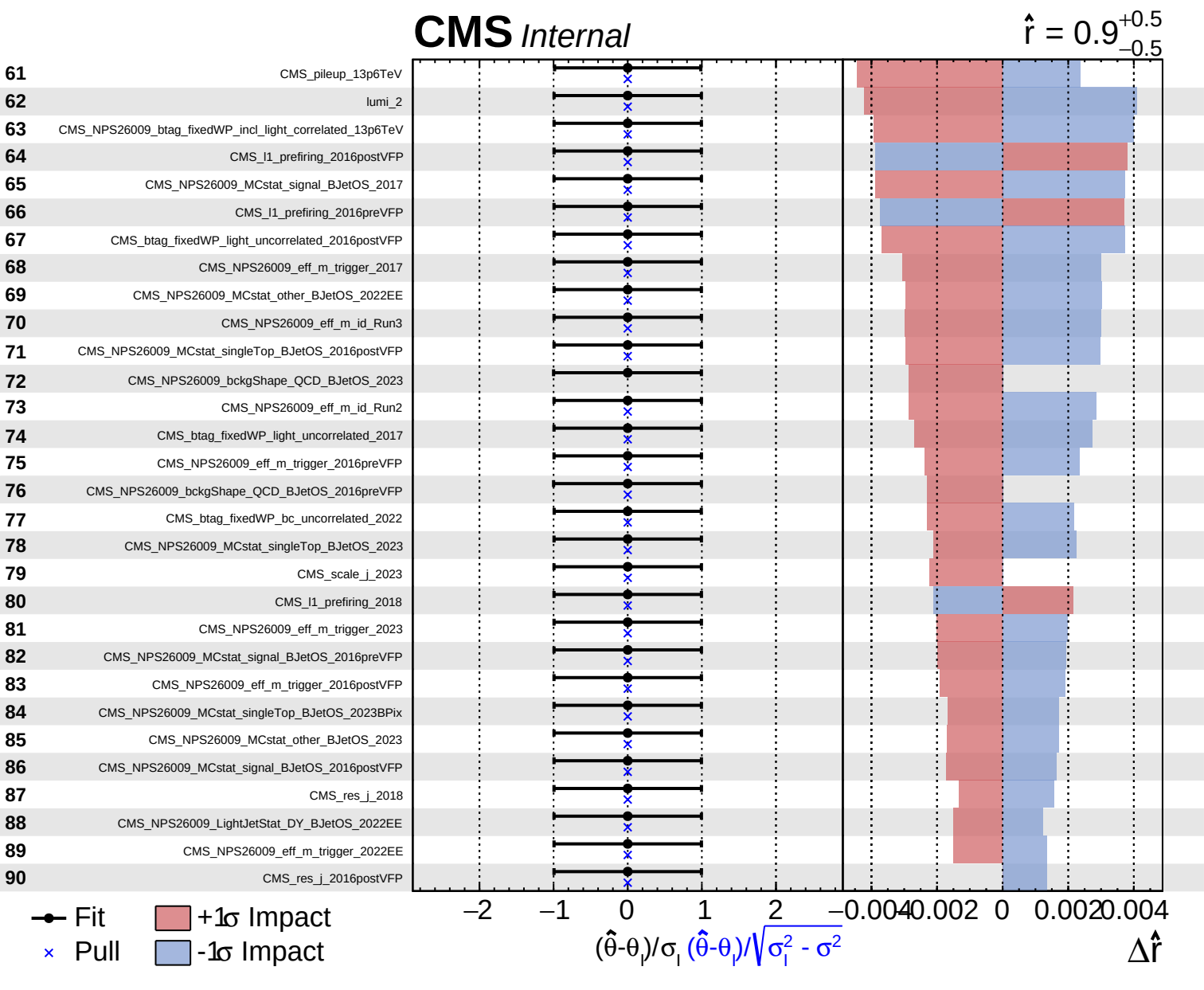
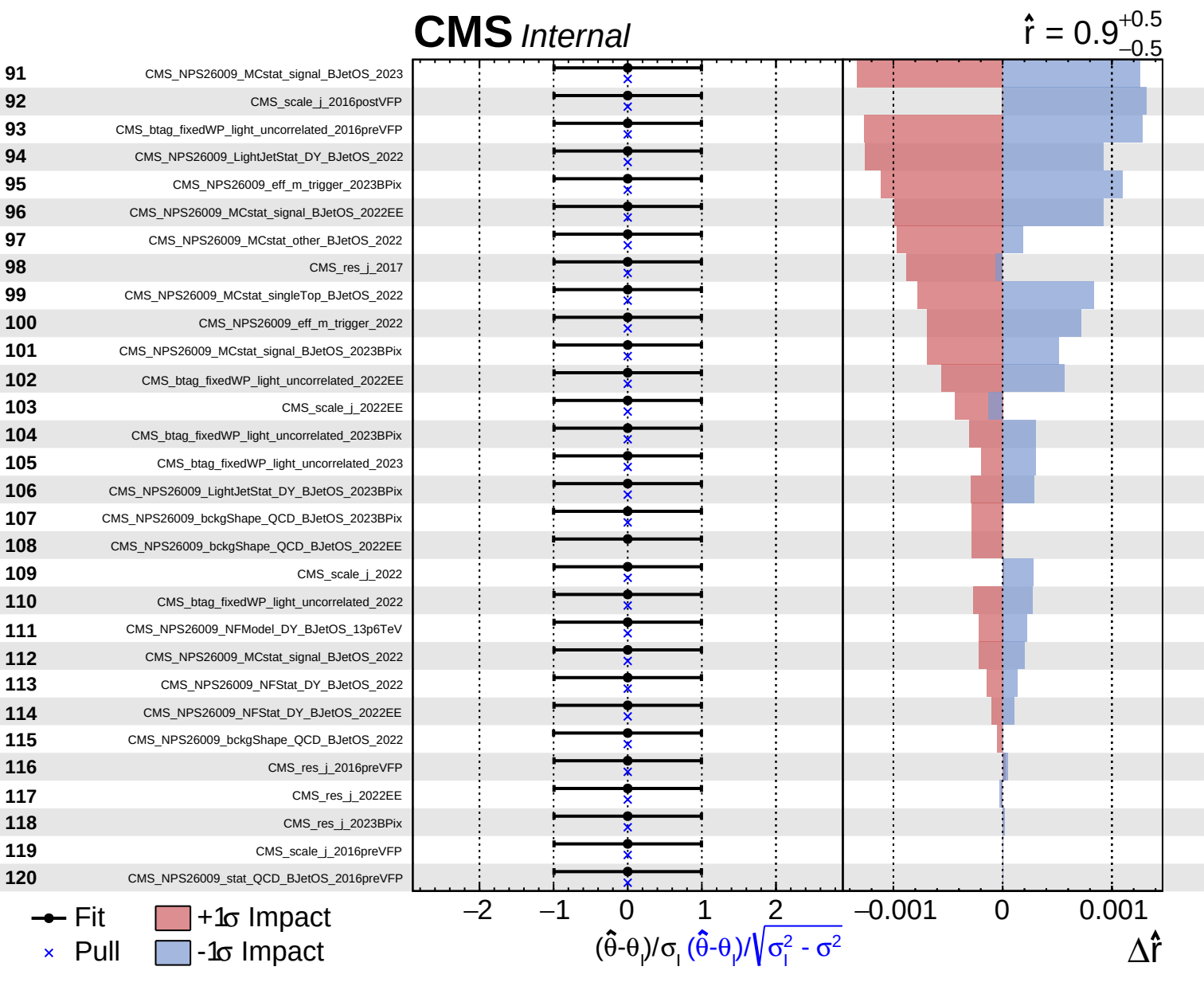
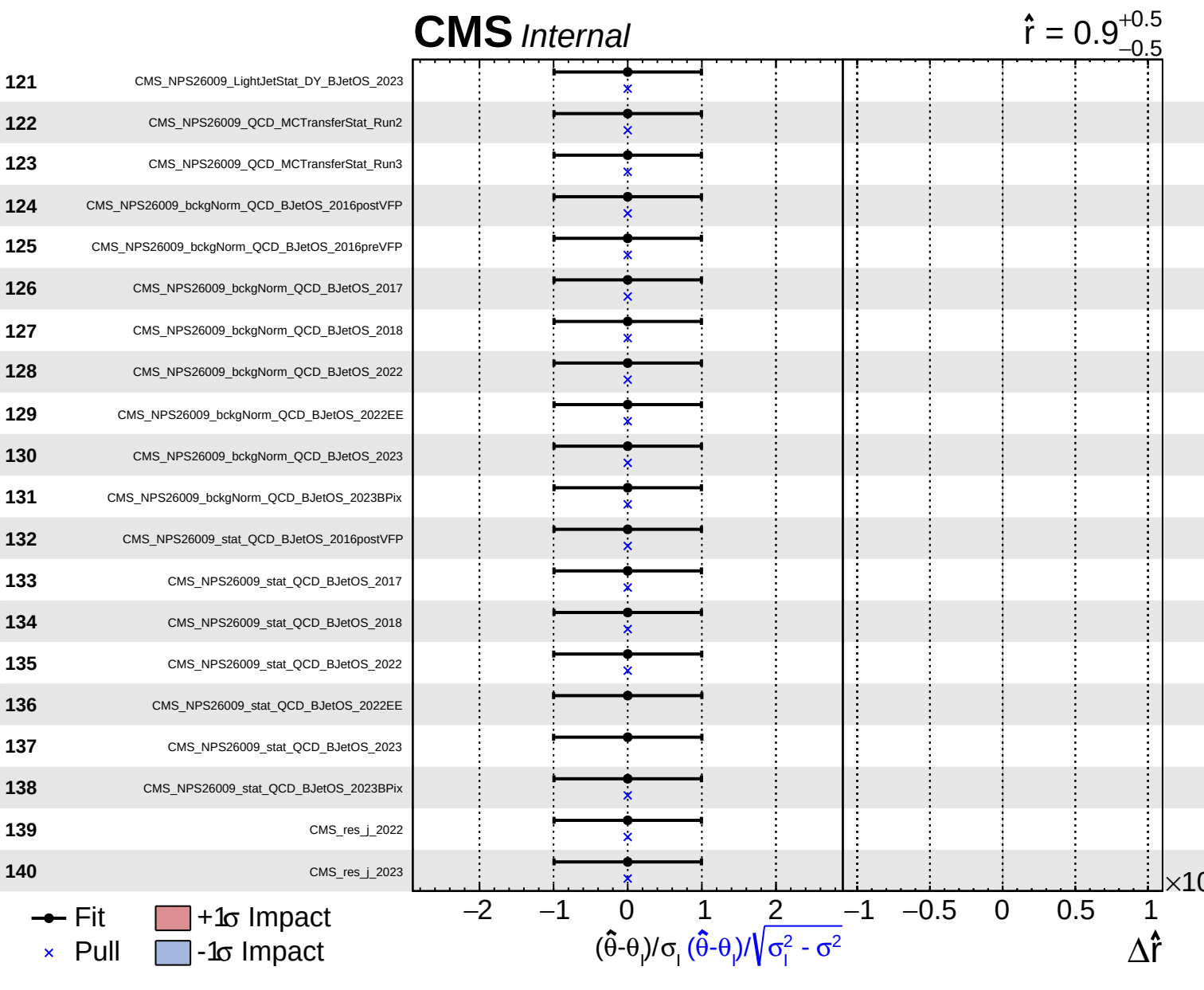

CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
1 QCDscale_ren_ttbar
2 QCDscale_fac_ttbar
3 CMS_NPS26009_NFModel_DY_BJetOS_13TeV
4 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13TeV
5 CMS_NPS26009_topmass_ttbar_BJetOS
6 CMS_NPS26009_pdf_NNPDF31
7 pdf_alphas
8 CMS_NPS26009_LightJetStat_DY_BJetOS_2018
9 CMS_NPS26009_NFStat_DY_BJetOS_2018
10 CMS_NPS26009_btag_fixedWP_comb_bc_correlated_13p6TeV
11 CMS_NPS26009_MCstat_ttbar_BJetOS_2018
12 CMS_NPS26009_LightJetStat_DY_BJetOS_2017
13 CMS_btag_fixedWP_bc_uncorrelated_2018
14 CMS_NPS26009_LightJetStat_DY_BJetOS_2016preVFP
15 CMS_NPS26009_MCstat_ttbar_BJetOS_2017
16 CMS_btag_fixedWP_bc_uncorrelated_2017
17 CMS_NPS26009_NFStat_DY_BJetOS_2017
18 CMS_NPS26009_MCstat_ttbar_BJetOS_2016preVFP
19 CMS_NPS26009_MCstat_ttbar_BJetOS_2023
20 CMS_NPS26009_MCstat_ttbar_BJetOS_2016postVFP
21 CMS_NPS26009_LightJetStat_DY_BJetOS_2016postVFP
22 CMS_NPS26009_MCstat_ttbar_BJetOS_2022EE
23 CMS_NPS26009_MCstat_ttbar_BJetOS_2023BPix
24 CMS_NPS26009_NFStat_DY_BJetOS_2016preVFP
25 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13TeV
26 CMS_NPS26009_bckgShape_QCD_BJetOS_2017
27 lumi_13TeV_15161718_I
28 CMS_NPS26009_bckgShape_QCD_BJetOS_2018
29 CMS_scale_m_13TeV
30 CMS_NPS26009_MCstat_singleTop_BJetOS_2022EE
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
−0.2 0 0.2
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
31 CMS_NPS26009_QCDscale_ren_singleTop
32 CMS_btag_fixedWP_bc_uncorrelated_2016preVFP
33 CMS_NPS26009_MCstat_singleTop_BJetOS_2018
34 CMS_NPS26009_MCstat_other_BJetOS_2018
35 CMS_NPS26009_NFStat_DY_BJetOS_2016postVFP
36 CMS_btag_fixedWP_bc_uncorrelated_2016postVFP
37 CMS_NPS26009_MCstat_other_BJetOS_2016preVFP
38 CMS_NPS26009_MCstat_other_BJetOS_2023BPix
39 CMS_pileup_13TeV
40 CMS_NPS26009_MCstat_other_BJetOS_2017
41 CMS_NPS26009_MCstat_other_BJetOS_2016postVFP
42 CMS_NPS26009_MCstat_ttbar_BJetOS_2022
43 CMS_NPS26009_MCstat_singleTop_BJetOS_2017
44 lumi_13TeV_1516_l
45 CMS_NPS26009_MCstat_singleTop_BJetOS_2016preVFP
46 CMS_NPS26009_QCDscale_fac_singleTop
47 CMS_NPS26009_bckgShape_QCD_BJetOS_2016postVFP
48 CMS_btag_fixedWP_bc_uncorrelated_2023
49 CMS_l1_prefiring_2017
50 CMS_scale_j_2023BPix
51 CMS_scale_m_13p6TeV
52 CMS_NPS26009_MCstat_signal_BJetOS_2018
53 CMS_btag_fixedWP_bc_uncorrelated_2022EE
54 CMS_btag_fixedWP_light_uncorrelated_2018
55 CMS_scale_j_2017
56 CMS_scale_j_2018
57 CMS_btag_fixedWP_bc_uncorrelated_2023BPix
58 lumi_13TeV_151617_l
59 CMS_NPS26009_eff_m_trigger_2018
60 lumi_1
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.01
0
0.01
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
r̂ = 0.9 +0.5 −0.5
61 CMS_pileup_13p6TeV
62 lumi_2
63 CMS_NPS26009_btag_fixedWP_incl_light_correlated_13p6TeV
64 CMS_l1_prefiring_2016postVFP
65 CMS_NPS26009_MCstat_signal_BJetOS_2017
66 CMS_l1_prefiring_2016preVFP
67 CMS_btag_fixedWP_light_uncorrelated_2016postVFP
68 CMS_NPS26009_eff_m_trigger_2017
69 CMS_NPS26009_MCstat_other_BJetOS_2022EE
70 CMS_NPS26009_eff_m_id_Run3
71 CMS_NPS26009_MCstat_singleTop_BJetOS_2016postVFP
72 CMS_NPS26009_bckgShape_QCD_BJetOS_2023
73 CMS_NPS26009_eff_m_id_Run2
74 CMS_btag_fixedWP_light_uncorrelated_2017
75 CMS_NPS26009_eff_m_trigger_2016preVFP
76 CMS_NPS26009_bckgShape_QCD_BJetOS_2016preVFP
77 CMS_btag_fixedWP_bc_uncorrelated_2022
78 CMS_NPS26009_MCstat_singleTop_BJetOS_2023
79 CMS_scale_j_2023
80 CMS_l1_prefiring_2018
81 CMS_NPS26009_eff_m_trigger_2023
82 CMS_NPS26009_MCstat_signal_BJetOS_2016preVFP
83 CMS_NPS26009_eff_m_trigger_2016postVFP
84 CMS_NPS26009_MCstat_singleTop_BJetOS_2023BPix
85 CMS_NPS26009_MCstat_other_BJetOS_2023
86 CMS_NPS26009_MCstat_signal_BJetOS_2016postVFP
87 CMS_res_j_2018
88 CMS_NPS26009_LightJetStat_DY_BJetOS_2022EE
89 CMS_NPS26009_eff_m_trigger_2022EE
90 CMS_res_j_2016postVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2 −1 0 1 2
(θ̂-θ_I)/σ_I
−0.004 −0.002 0 0.002 0.004
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂


CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
91 CMS_NPS26009_MCstat_signal_BJetOS_2023
92 CMS_scale_j_2016postVFP
93 CMS_btag_fixedWP_light_uncorrelated_2016preVFP
94 CMS_NPS26009_LightJetStat_DY_BJetOS_2022
95 CMS_NPS26009_eff_m_trigger_2023BPix
96 CMS_NPS26009_MCstat_signal_BJetOS_2022EE
97 CMS_NPS26009_MCstat_other_BJetOS_2022
98 CMS_res_j_2017
99 CMS_NPS26009_MCstat_singleTop_BJetOS_2022
100 CMS_NPS26009_eff_m_trigger_2022
101 CMS_NPS26009_MCstat_signal_BJetOS_2023BPix
102 CMS_btag_fixedWP_light_uncorrelated_2022EE
103 CMS_scale_j_2022EE
104 CMS_btag_fixedWP_light_uncorrelated_2023BPix
105 CMS_btag_fixedWP_light_uncorrelated_2023
106 CMS_NPS26009_LightJetStat_DY_BJetOS_2023BPix
107 CMS_NPS26009_bckgShape_QCD_BJetOS_2023BPix
108 CMS_NPS26009_bckgShape_QCD_BJetOS_2022EE
109 CMS_scale_j_2022
110 CMS_btag_fixedWP_light_uncorrelated_2022
111 CMS_NPS26009_NFModel_DY_BJetOS_13p6TeV
112 CMS_NPS26009_MCstat_signal_BJetOS_2022
113 CMS_NPS26009_NFStat_DY_BJetOS_2022
114 CMS_NPS26009_NFStat_DY_BJetOS_2022EE
115 CMS_NPS26009_bckgShape_QCD_BJetOS_2022
116 CMS_res_j_2016preVFP
117 CMS_res_j_2022EE
118 CMS_res_j_2023BPix
119 CMS_scale_j_2016preVFP
120 CMS_NPS26009_stat_QCD_BJetOS_2016preVFP
Fit
Pull
+1σ Impact
-1σ Impact
−2
−1
0
1
2
$(\hat{\theta}-\theta_I)/\sigma_I$
−0.001
0
0.001
$(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$


CMS Internal
$\hat{r} = 0.9^{+0.5}_{-0.5}$
121 CMS_NPS26009_LightJetStat_DY_BJetOS_2023
122 CMS_NPS26009_QCD_MCTransferStat_Run2
123 CMS_NPS26009_QCD_MCTransferStat_Run3
124 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016postVFP
125 CMS_NPS26009_bckgNorm_QCD_BJetOS_2016preVFP
126 CMS_NPS26009_bckgNorm_QCD_BJetOS_2017
127 CMS_NPS26009_bckgNorm_QCD_BJetOS_2018
128 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022
129 CMS_NPS26009_bckgNorm_QCD_BJetOS_2022EE
130 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023
131 CMS_NPS26009_bckgNorm_QCD_BJetOS_2023BPix
132 CMS_NPS26009_stat_QCD_BJetOS_2016postVFP
133 CMS_NPS26009_stat_QCD_BJetOS_2017
134 CMS_NPS26009_stat_QCD_BJetOS_2018
135 CMS_NPS26009_stat_QCD_BJetOS_2022
136 CMS_NPS26009_stat_QCD_BJetOS_2022EE
137 CMS_NPS26009_stat_QCD_BJetOS_2023
138 CMS_NPS26009_stat_QCD_BJetOS_2023BPix
139 CMS_res_j_2022
140 CMS_res_j_2023
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Fit
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_I)/\sigma_I$ $(\hat{\theta}-\theta_I)/\sqrt{\sigma_I^2 - \sigma^2}$
$\Delta\hat{r}$